

Celebal Excellence Internship 2026

Week 4 - Azure Cloud Fundamentals and Data Pipeline Implementation using
ADF

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Dataset: Sample - Superstore.csv

Objective

To understand Azure cloud concepts and build a complete end-to-end data pipeline using Azure Storage Account and Azure Data Factory (ADF). The assignment covers resource provisioning, pipeline design, metadata validation, IAM role assignment, and pipeline execution monitoring.

Task 1 - Azure Portal & Resource Group

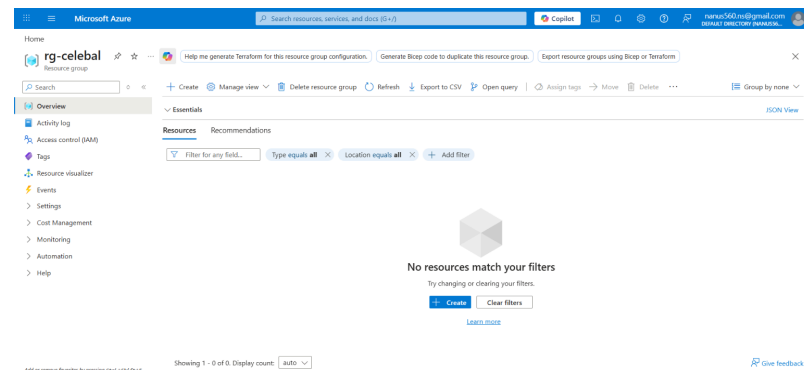
Steps Performed

- Logged into Azure Portal at portal.azure.com
- Navigated to Resource Groups via the search bar
- Clicked + Create and filled in the following configuration:

Subscription	Azure for Students
Resource Group Name	rg-celebal
Region	Central India

- Clicked Review + Create → Create
- Verified Resource Group appeared in the list after deployment

Screenshot-



Task 2 - Storage Account Setup

Steps Performed

2a. Create Storage Account

- Searched for Storage Accounts in Azure Portal → clicked + Create

Resource Group	rg-celebal-week4
Storage Account Name	celebalweek4storage
Region	Central India

Performance	Standard
Redundancy	LRS (Locally Redundant Storage)

Screenshot-

Home

celebalweek4storage Show me alerts for this storage account Enhance the security of this storage account Assess data resiliency for this storage account

Storage account

Upload Open in Explorer Delete Move Refresh Open in mobile CLI / PS Feedback

Essentials [JSON View](#)

Resource group (new)	: rg-celebal	Performance	: Standard
Location	: centralindia	Replication	: Locally redundant storage (LRS)
Subscription (new)	: Azure for Students	Account kind	: StorageV2 (general purpose v2)
Subscription ID	: 76687b95-eb63-451a-b696-c697d6e658cd	Provisioning state	: Succeeded
Disk state	: Available	Created	: 6/14/2026, 9:54:49 PM
Tags (edit)	: Add tags		

Properties Monitoring Capabilities (7) Recommendations (0) Tutorials Tools + SDKs

Blob service

Hierarchical namespace	Disabled
Default access tier	Hot
Blob anonymous access	Disabled
Blob soft delete	Enabled (7 days)
Container soft delete	Enabled (7 days)
Versioning	Disabled
Change feed	Disabled

Security

Require secure transfer for REST API operations	Enabled
Storage account key access	Enabled
Minimum TLS version	Version 1.2
Infrastructure encryption	Disabled

Networking

Public network access	Enabled
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2b. Create Blob Containers

- Inside the Storage Account → navigated to Containers → + Container
- Created container: superstore-input (for source CSV)
- Created container: superstore-output (for processed/destination file)
- Access level kept as Private (no anonymous access)

Screenshot- (superstore-input)

Microsoft Azure

superstore-input

Container

Search

Overview

Authentication method: Access key (switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Name	Last modified	Access tier	Blob type	Size	Lease state
No items found					

Add or remove favorites by pressing ctrl + shift + F

Screenshot- (superstore-output)

Home > celebalweek4storage | Containers

superstore-output

Container

Search

Overview

Authentication method: Access key (switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

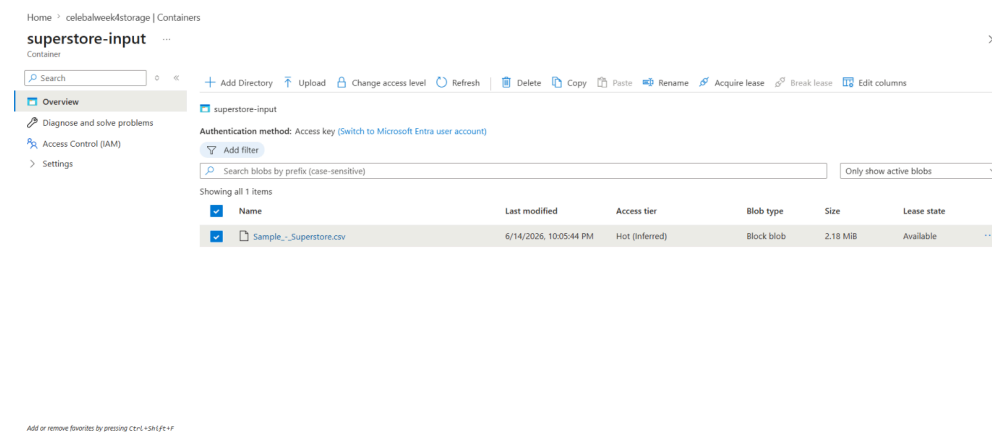
Name	Last modified	Access tier	Blob type	Size	Lease state
No items found					

Add or remove favorites by pressing ctrl + shift + F

2c. Upload CSV File

- Opened superstore-input container → clicked Upload
- Selected and uploaded: Sample_-_Superstore.csv
- Verified file appeared with correct size and last-modified timestamp

Screenshot-



Task 3 - Azure Data Factory Setup

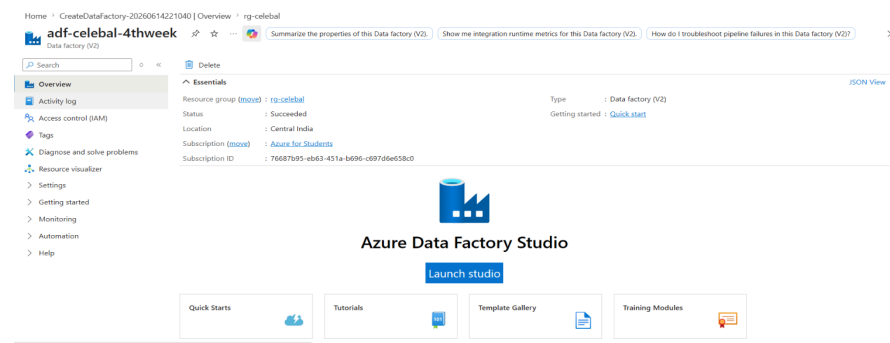
3a. Create Azure Data Factory

- Searched for Data Factories in Azure Portal → clicked + Create

Resource Group	rg-celebal
ADF Name	adf-celebal-4thweek
Region	Central India
Version	V2

- Clicked Review + Create → Create (deployment took ~2-3 minutes)
- After deployment, clicked Launch Studio to open ADF UI

Screenshot-

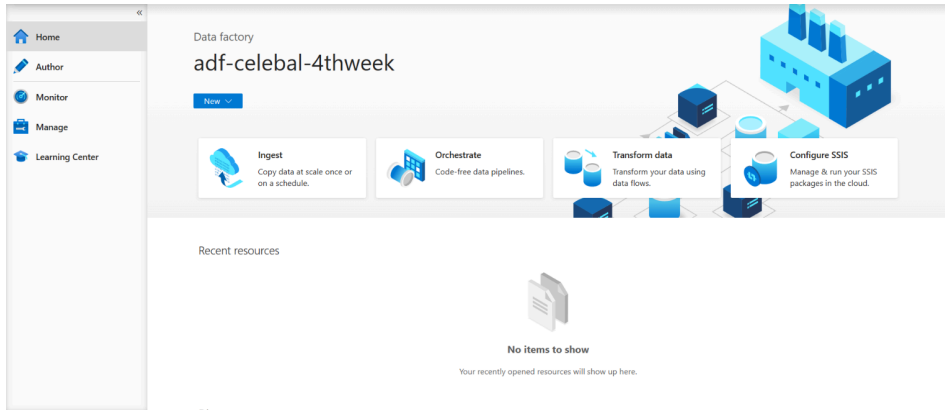


3b. ADF UI Exploration

The ADF Studio consists of three main sections:

Tab	Purpose
Author	Create/edit pipelines, datasets, linked services
Monitor	View pipeline execution status, debug logs, trigger runs
Manage	Manage linked services, integration runtimes, Git config

Screenshot-



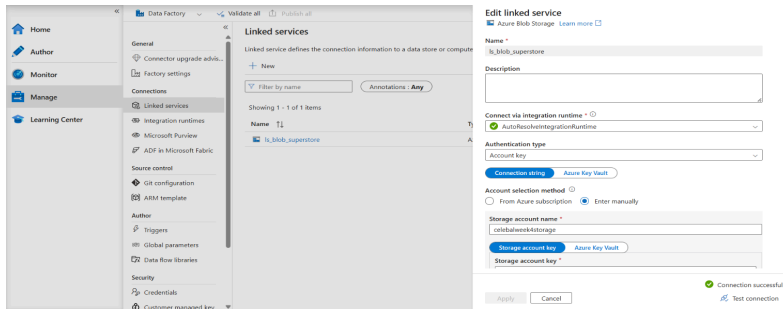
3c. Create Linked Service

- Navigated to Manage tab → Linked Services → + New
- Selected Azure Blob Storage → Continue

Linked Service Name	Is_blob_superstore
Storage Account	celebalweek4storage
Authentication	Account Key (auto-populated)

- Clicked Test Connection → received Connection Successful confirmation
- Clicked Create to save

Screenshot-



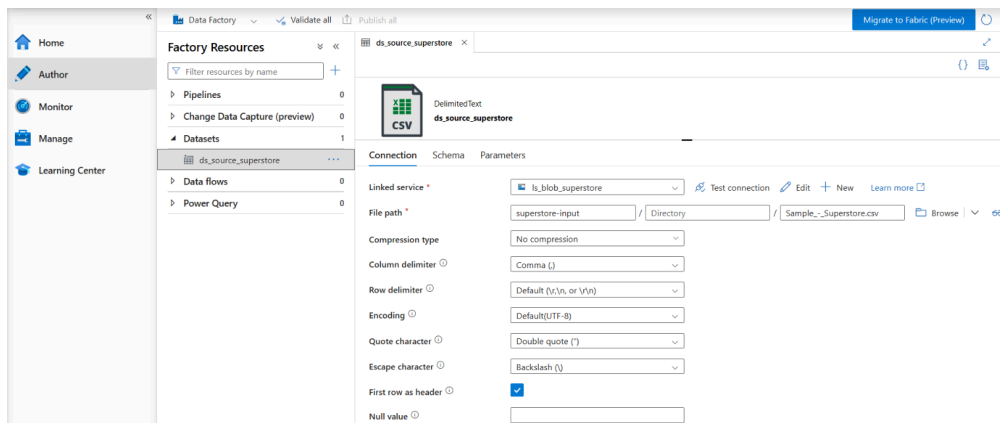
3d. Create Datasets

Source Dataset

- Author tab → Datasets → New Dataset → Azure Blob Storage → DelimitedText

Dataset Name	ds_source_superstore
Linked Service	ls_blob_superstore
Container	superstore-input
File	Sample_-_Superstore.csv
First Row as Header	Yes

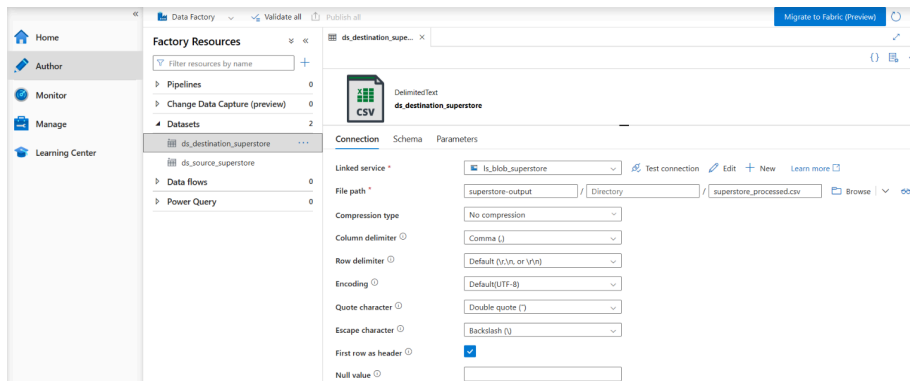
Screenshot-



Destination Dataset

Dataset Name	ds_destination_superstore
Linked Service	ls_blob_superstore
Container	superstore-output
File	superstore_processed.csv
First Row as Header	Yes

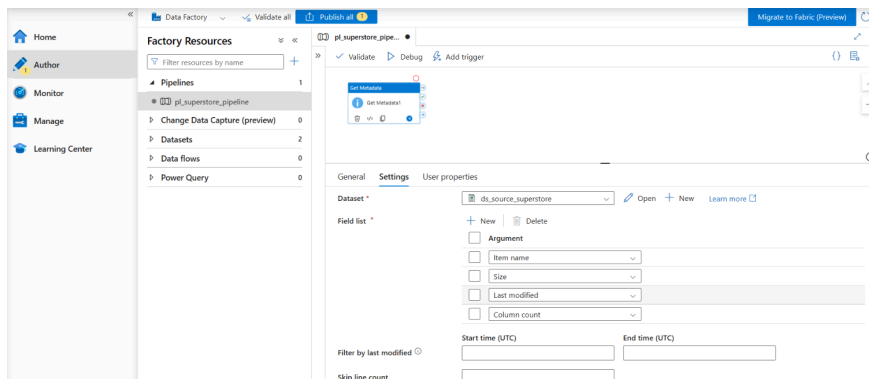
Screenshot-



3e. Get Metadata Activity

- Created a new pipeline: pl_superstore_pipeline
- Dragged Get Metadata activity from the Activities panel onto the canvas
- In Settings tab, selected dataset: ds_source_superstore
- Added the following field list items:
- itemName — retrieves file name
- size — retrieves file size in bytes
- lastModified — retrieves last modification timestamp
- columnCount — retrieves number of columns in the file

Screenshot-



Task 4 - Pipeline Development with Copy Data Activity

Steps Performed

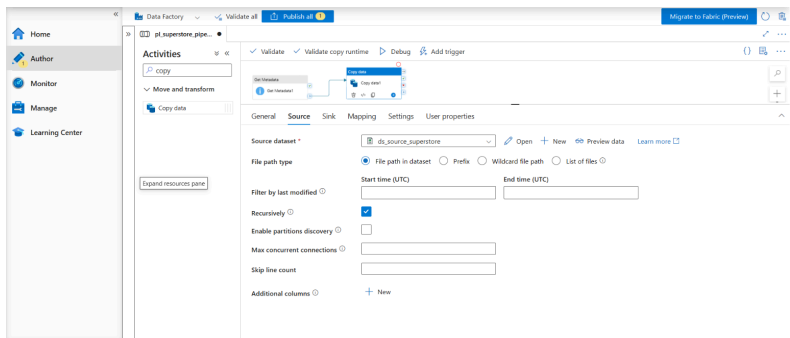
- In the same pipeline pl_superstore_pipeline, dragged Copy Data activity onto the canvas
- Connected Get Metadata → Copy Data using the green success arrow
- This ensures metadata is validated before copying begins

Copy Data Configuration

Source Tab

Dataset	ds_source_superstore
---------	----------------------

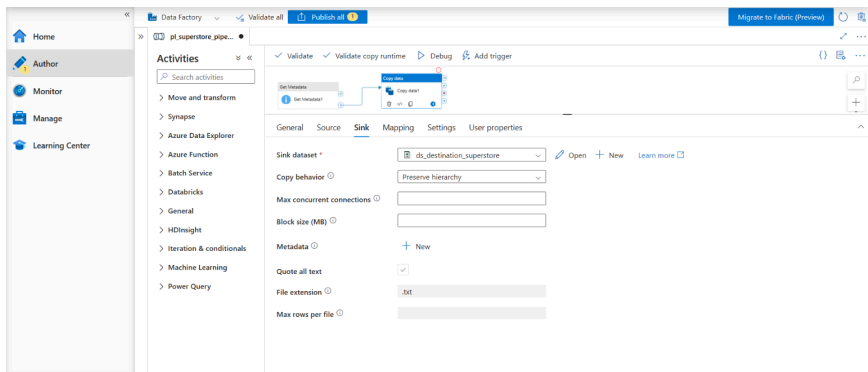
Screenshot-



Sink Tab

Dataset	ds_destination_superstore
Copy Behavior	PreserveHierarchy

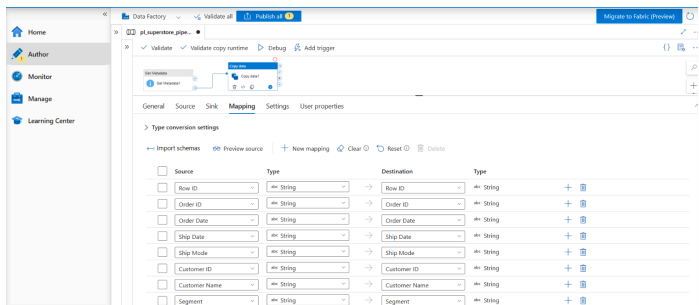
Screenshot-



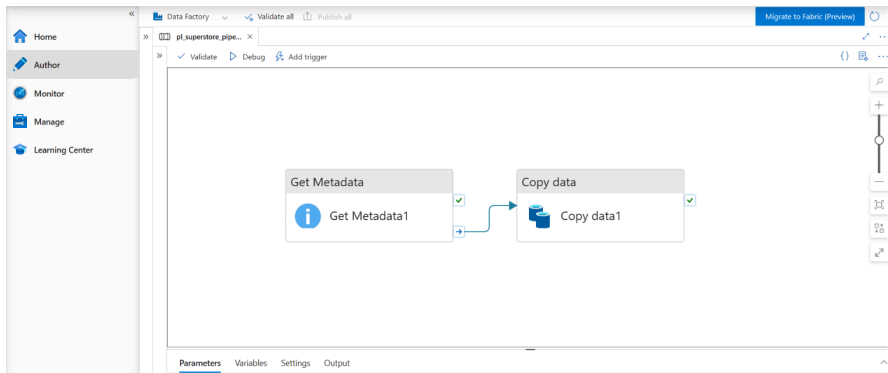
Mapping Tab

- Clicked Import Schemas - all 21 columns from Superstore CSV auto-mapped
- Verified column mapping: Row ID, Order ID, Order Date, Ship Date, ... Profit

Screenshot-



Pipeline Architecture



Task 5 - Pipeline Execution

Debug Run

- Clicked Debug button in the pipeline toolbar to run without publishing
- Monitored real-time execution in the Output tab at the bottom.

Screenshot-

This screenshot shows the 'Output' tab of the pipeline execution. The pipeline status is 'Succeeded'. Below the status, there is a table with 10 columns: 'Activity status', 'Activity name', 'Run start', 'Duration', 'Integration runtime', 'User properties', 'Activity run ID', and 'Log'. The table contains two rows of data, both marked as 'Succeeded'.

Activity status	Activity name	Run start	Duration	Integration runtime	User properties	Activity run ID	Log
Succeeded	Copy data	6/14/2026, 11:13:58 PM	17s	AutoResolveIntegrationRuntime (Central India)		dd5e17b7-4e1b-4545-ba6c-79d2b0129de	
Succeeded	Get Metadata	6/14/2026, 11:13:41 PM	16s	AutoResolveIntegrationRuntime (Central India)		65fb04e1-5c30-4a08-8342-ac15b8f5442	

Activity	Status	Result
Get Metadata	Succeeded	Returned itemName, size, lastModified, columnCount
Copy Data	Succeeded	Rows read & written: 9994 Duration: 17s

Output Verification

- Navigated to Storage Account → superstore-output container
- Confirmed file superstore_processed.csv was created
- Verified file size matched the source CSV

Screenshot-(Get Metadata Output)

The screenshot shows the 'Get Metadata' activity in the pipeline. The output window displays the following JSON data:

```
{
  "itemName": "Sample_Superstore.csv",
  "size": 2287806,
  "lastModified": "2026-06-14T16:35:44Z",
  "columnCount": 21,
  "effectiveIntegrationRuntime": "AutoResolveIntegrationRuntime (Central India)",
  "executionDuration": 1,
  "durationInQueue": 1
}
```

The pipeline status is 'Succeeded'. The table below shows the execution details:

Run start	Duration	Integration runtime	User prop...	Activity run ID
6/14/2026, 11:13:58 PM	17s	AutoResolveIntegrationRuntime (Central India)		ddee17b7-4e18-4545-ba6c-79d2bb0129de
6/14/2026, 11:13:41 PM	16s	AutoResolveIntegrationRuntime (Central India)		65b04e1-5c30-4a0d-8342-ac15b8f5442

Screenshot-(Copy Data Output)

The screenshot shows the 'Copy data' activity in the pipeline. The output window displays the following JSON data:

```
{
  "dataRead": 2287806,
  "dataWritten": 2616204,
  "filesRead": 1,
  "filesWritten": 1,
  "sourcePeakConnections": 1,
  "sinkPeakConnections": 1,
  "rowsRead": 9994,
  "rowsCopied": 9994
}
```

The pipeline status is 'Succeeded'. The table below shows the execution details:

Run start	Duration	Integration runtime	User prop...	Activity run ID
6/14/2026, 11:13:58 PM	17s	AutoResolveIntegrationRuntime (Central India)		ddee17b7-4e18-4545-ba6c-79d2bb0129de
6/14/2026, 11:13:41 PM	16s	AutoResolveIntegrationRuntime (Central India)		65b04e1-5c30-4a0d-8342-ac15b8f5442

Screenshot- (Data at Destination)

The screenshot shows the 'superstore-output' container in Azure Storage Explorer. The 'Overview' tab displays the following information:

- Authentication method: Access key (Switch to Microsoft Entra user account)
- Search blobs by prefix (case-sensitive):
- Showing all 1 items

Name	Last modified	Access tier	Blob type	Size	Lease state
superstore_processed.csv	6/14/2026, 11:14:12 PM	Hot (preferred)	Block blob	2.5 MiB	Available

Task 6 - IAM Role Assignment

Steps Performed

- Navigated to Storage Account (celebalweek4storage) → Access Control (IAM)
- Clicked + Add → Add role assignment for each of the following:

Role Assignments

Role	Assigned To	Purpose
Reader	Saksham (own account)	Read-only access to storage resources
Storage Account Contributor	Saksham (own account)	Create and manage storage resources
Storage Blob Data Contributor	adf-celebal-week4 (Managed Identity)	ADF can read/write to Blob Storage

Screenshot-(Reader)

Home > celebalweek4storage

celebalweek4storage | Access Control (IAM) ☆ --

Storage account

+ Add ▾ Download role assignments Edit columns Refresh Delete Feedback

Check access **Role assignments** Roles Deny assignments

Number of role assignments for this subscription 2

Search by name or email Type: All Role: All Scope: All scopes Group by: Role

All (2) Job function roles (1) Privileged administrator roles (1)

Name ↑	Type ↑	Role ↑	Scope ↑	Condition ↑
> Owner (1)				
> Reader (1)				
 Saksham Sharma 54546bd8-1bea-43dc-af04-befedfeb...	User	Reader	This resource	None

Showing 1 - 2 of 2 results.

Screenshot-(Storage Account Contributor)

Home > celebalweek4storage

celebalweek4storage | Access Control (IAM) ☆ --

Storage account

+ Add ▾ Download role assignments Edit columns Refresh Delete Feedback

Check access **Role assignments** Roles Deny assignments

Number of role assignments for this subscription 3

Search by name or email Type: All Role: All Scope: All scopes Group by: Role

All (3) Job function roles (2) Privileged administrator roles (1)

Name ↑	Type ↑	Role ↑	Scope ↑	Condition ↑
> Owner (1)				
> Reader (1)				
> Storage Account Contributor (1)				
 Saksham Sharma 54546bd8-1bea-43dc-af04-befedfeb...	User	Storage Account Contributor	This resource	None

Showing 1 - 3 of 3 results.

Screenshot-(Get Metadata Output)

Home > celebalweek4storage

celebalweek4storage | Access Control (IAM) ☆ --

Storage account

+ Add ▾ Download role assignments Edit columns Refresh Delete Feedback

Check access **Role assignments** Roles Deny assignments

Number of role assignments for this subscription 4

Search by name or email Type: All Role: All Scope: All scopes Group by: Role

All (4) Job function roles (3) Privileged administrator roles (1)

Name ↑	Type ↑	Role ↑	Scope ↑	Condition ↑
> Owner (1)				
> Reader (1)				
> Storage Account Contributor (1)				
> Storage Blob Data Contributor (1)				
 adf-celebal-4fweek4 55599f9f-4a75-4c5f-bd4e-827a633b...	Managed identity	Storage Blob Data Contributor	This resource	Add

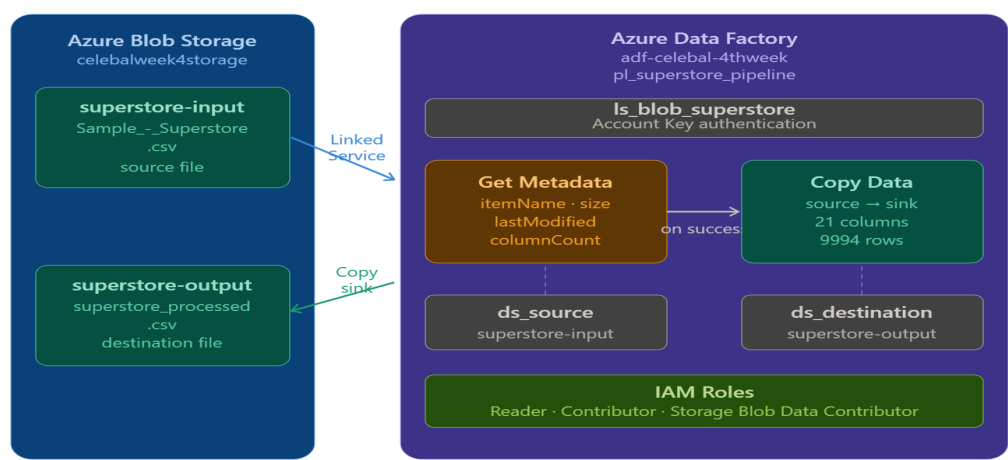
Showing 1 - 4 of 4 results.

Mini Project - End-to-End Pipeline

Problem Statement

Build a complete pipeline that reads a CSV file from Azure Blob Storage and processes it using Azure Data Factory, copying data from source to destination with metadata validation.

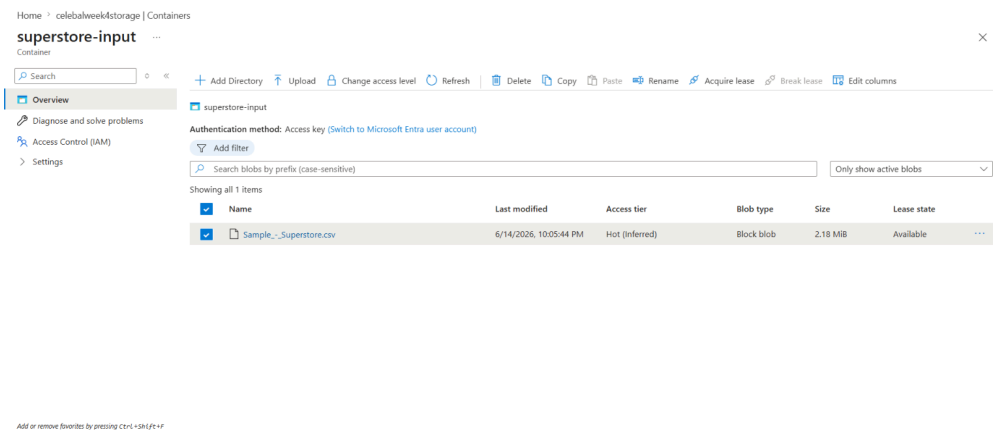
Architecture Overview



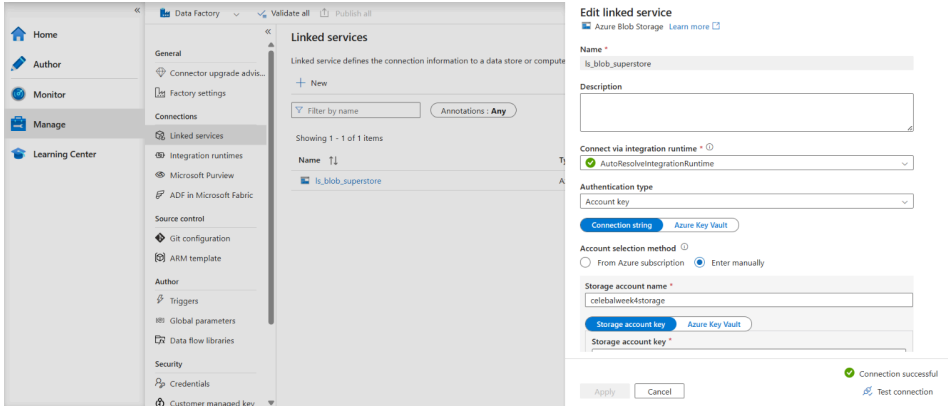
Requirements Fulfilled

Requirement	Implementation
Source: CSV in Blob	Sample_-_Superstore.csv in superstore-input container
Linked Service	Is_blob_superstore (Azure Blob Storage)
Source Dataset	ds_source_superstore (DelimitedText, 21 columns)
Destination Dataset	ds_destination_superstore (superstore-output)
Metadata Check	Get Metadata activity — itemName, size, lastModified, columnCount
Copy Data Activity	Full 21-column copy with schema mapping
Pipeline Executed	Debug run - both activities Succeeded
Data at Destination	superstore_processed.csv created in superstore-output

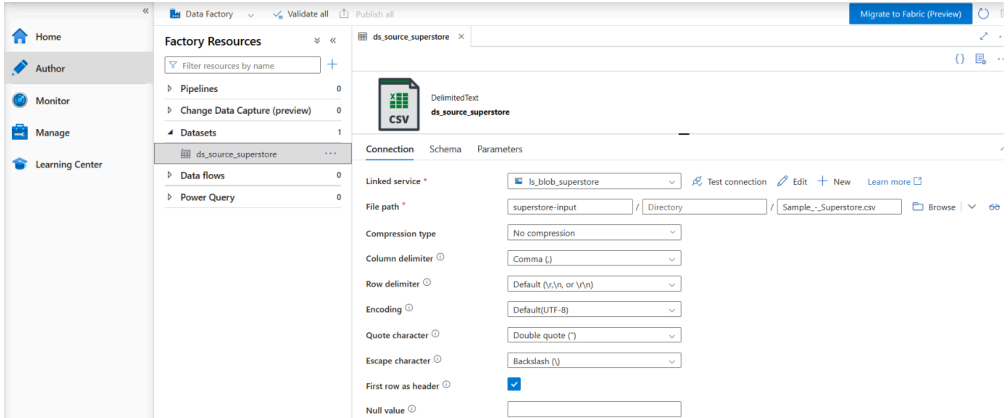
Screenshot-(Source: CSV in Blob)



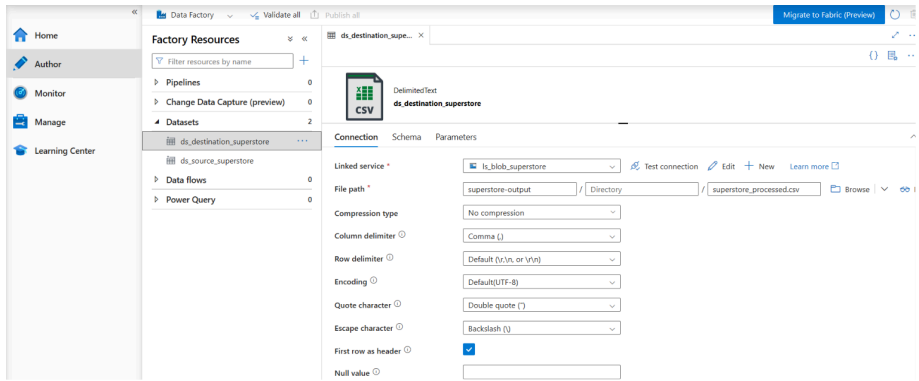
Screenshot-(Linked Services)



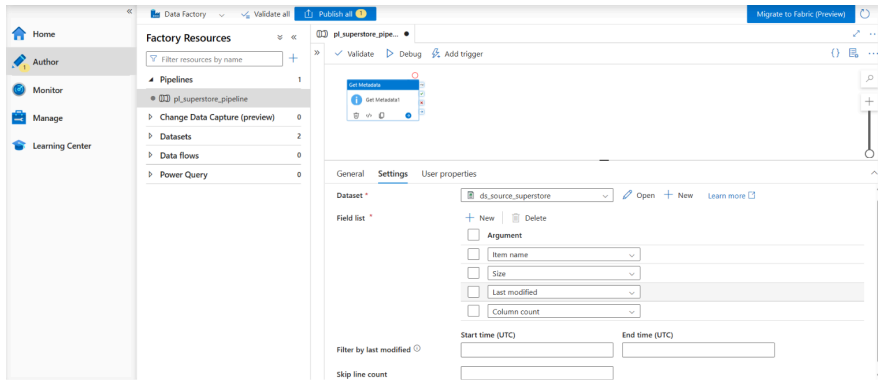
Screenshot-(Source Dataset)



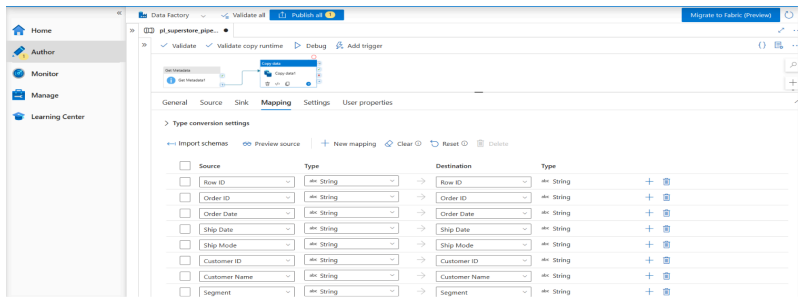
Screenshot-(Destination Dataset)



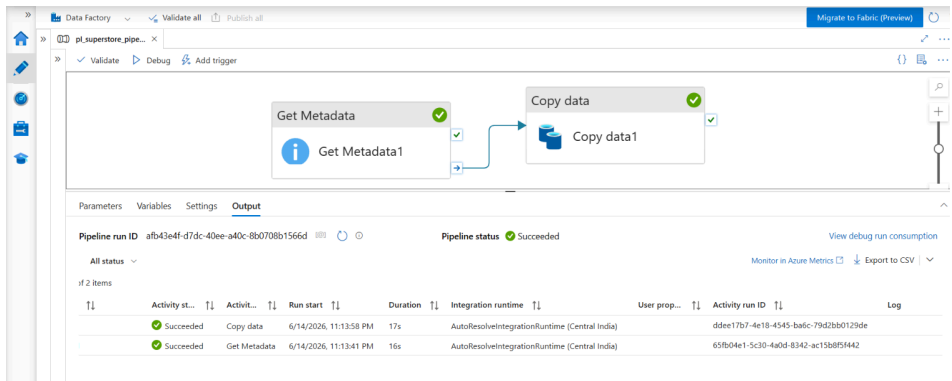
Screenshot-(Metadata Check)



Screenshot-(Copydata Activity)



Screenshot-(Pipeline Executed)



Screenshot-(Data at Destination)

Home > celebratweek4storage | Containers

superstore-output

Container

+ Add Directory ↑ Upload ↻ Change access level ↺ Refresh 🗑️ Delete 📋 Copy 📄 Paste 🏷️ Rename 🔑 Acquire lease 🔑 Break lease ⚙️ Edit columns

📁 Overview
📁 superstore-output

🔧 Diagnose and solve problems

🔑 Access Control (IAM)

➤ Settings

Authentication method: Access key ([Switch to Microsoft Entra user account](#))

🔍 Add filter

🔍 Search blobs by prefix (case-sensitive) Only show active blobs

Showing all 1 items

☒	Name	Last modified	Access tier	Blob type	Size	Lease state
☒	📄 superstore_processed.csv	6/14/2026, 11:14:12 PM	Hot (inferred)	Block blob	2.5 MiB	Available

Add or remove favorites by pressing Ctrl+Shift+F

Summary

This assignment provided hands-on experience with Azure cloud services and the Azure Data Factory pipeline ecosystem. Key learnings:

- Set up the Azure environment from scratch by creating a Resource Group, Storage Account, and two Blob Containers (superstore-input and superstore-output) through the Azure Portal.
- Explored the Azure Data Factory Studio and its three core sections: Author (pipeline design), Monitor (run tracking), and Manage (linked services and configuration).
- Configured a Linked Service (ls_blob_superstore) to establish a secure connection between ADF and the Azure Blob Storage account using Account Key authentication.
- Defined two datasets in DelimitedText (CSV) format: ds_source_superstore pointing to the uploaded Superstore file, and ds_destination_superstore for the pipeline output.
- Added a Get Metadata activity to the pipeline that retrieves key file attributes (itemName, size, lastModified, columnCount) from the source CSV before any data movement begins.
- Built the pipeline pl_superstore_pipeline by chaining Get Metadata and Copy Data activities with a success dependency arrow, ensuring metadata is checked before data is moved.
- Configured the Copy Data activity with source and sink datasets, imported the full 21-column schema from the Superstore CSV, and set fault tolerance to handle any malformed rows.
- Ran the pipeline using the Debug trigger and monitored the execution in the Output tab, with both Get Metadata and Copy Data activities completing successfully in under 35 seconds.
- Assigned Reader and Contributor roles to the user account via IAM, and granted the Storage Blob Data Contributor role to the ADF Managed Identity for secure, key-free access to Blob Storage.
- Verified the end-to-end pipeline output by checking the superstore-output container, where superstore_processed.csv was successfully created with all 9,994 rows copied from the source.